

No Easy Wins: A Contamination-Controlled Benchmark for Evaluating Anomaly Detection and Model Selection

Alexander Apartsin^a, Yehudit Apherstein^{b,*}

^a*School of Computer Science, Faculty of Sciences, Holon Institute of Technology (HIT), Holon, Israel*

^b*Intelligent Systems, Afeka Academic College of Engineering, Tel-Aviv, Israel*

Abstract

Generic anomaly detection methods are commonly evaluated on benchmarks that combine datasets containing normal observations and labeled anomalies. Designing such benchmarks is itself challenging. Existing datasets may contain duplicated anomalies, train/test leakage, anomalies that are trivially separable, or anomalies that are effectively indistinguishable from normal observations. Some also provide insufficient data for reliable comparisons, while commonly used evaluation metrics can be distorted by differences in anomaly prevalence. Reliable benchmarking therefore requires curated datasets, prevalence-aware evaluation, and statistically grounded comparisons between methods.

We introduce ADReal, a benchmark constructed by systematically curating and filtering datasets from multiple existing anomaly detection benchmarks. We remove duplicates, contaminated anomalies, leakage, and datasets that are either trivially easy or effectively impossible, yielding 151 datasets spanning tabular and multivariate time-series data. On these datasets, we evaluate roughly thirty widely used detectors, including classical isolation-, density-, distribution-, and projection-based methods and deep one-class and reconstruction-based models. Detectors are evaluated using a prevalence-aware metric that enables more comparable performance estimates across datasets with different anomaly rates.

ADReal also addresses a second problem: model selection. In real applications, anomalies are rare and heterogeneous, so a representative labeled anomaly sample may not be available for model selection or hyperparameter tuning. The detector must therefore often be selected using normal data alone. ADReal provides a controlled testbed for evaluating such normal-only selection methods: candidate detectors are selected without access to true anomalies, while their eventual performance on those anomalies provides an independent measure of selection quality. Using this testbed we evaluate popular unsupervised model-selection heuristics and release their results alongside the benchmark and detector evaluations.

Finally, we introduce SPARC, a normal-only model-selection method that generates synthetic anomalies from normal data and ranks candidate detectors by their ability to identify them. SPARC significantly outperforms the strongest competing selection heuristics. Its regret, the performance gap relative to the best available detector, is 0.124, compared with 0.176 for entropy and 0.174 for mass-volume ($p < 0.001$ against each). The advantage is particularly large for time-series data, where regret decreases to 0.05 versus 0.15 for the strongest baseline, and also holds on tabular data. We release ADReal, the complete detector evaluations, SPARC, and the associated code.

1. Introduction

New anomaly detection methods are introduced and compared on benchmarks that pair normal observations with labeled anomalies, drawn from fraud, industrial monitoring, health, and cybersecurity. The ranking such a benchmark produces is only as trustworthy as the data behind it. And because no single detector dominates, Isolation Forest wins on some datasets, LOF on others, ECOD on yet others, a practitioner must still choose one

detector per dataset, usually with no labeled anomalies to guide the choice. A decade of work has proposed label-free selection criteria for exactly this, from entropy and mass-volume curves [1] to internal relative-evaluation scores [2], meta-learning [3], and rank aggregation across a model pool [4]. Detectors and the selectors that choose among them are judged on the same benchmarks, so both inherit whatever those benchmarks get wrong.

Existing benchmarks are contaminated in ways that distort both kinds of comparison. Many anomalies are separable by a single feature or a one-line threshold rule, so every detector scores well and the choice between

*Corresponding author

Email address: apersteiny@afeka.ac.il (Yehudit Apherstein)

them barely matters. Some anomalies are exact copies of normal points, labeled both ways because the feature vector does not capture what makes them anomalous, and no detector can ever separate them. Normals leak between train and test when a dataset's duplicate rows are split at random, so a detector is scored on points it has already seen. Some tasks are effectively unsolvable, with no detector beating a random ranking, so a comparison on them carries no signal. Base rates drift so high that the scoring metric divides by a vanishing denominator. And differences are reported as point gaps with no test of significance. A benchmark built on such data measures artifacts, of the detectors and of the methods that choose among them.

We rebuild the benchmark and audit it until it holds. ADReal is a collection of tabular and multivariate time-series datasets produced by one detector-free pipeline whose construction rules each close one contamination channel. On the resulting 151 datasets we evaluate roughly thirty classical and deep detectors under a single consistent preprocessing, scored by a prevalence-aware metric, so no detector is handicapped by an inconsistency in the harness and the comparison is fair across datasets of different anomaly rates. ADReal then serves a second purpose: because a detector must in practice be chosen from normal data alone, the benchmark is also a controlled testbed for unsupervised model selection, where a method picks one detector without labels and is scored by its regret against the oracle-best detector, its criterion computed on held-out normals only. We evaluate eight popular selection methods under this protocol.

The honest protocol changes the result. Agreement-based selectors, which dominate under transductive evaluation, fall below a random pick once validation sees only normals; entropy and mass-volume survive. Building on this, we introduce SPARC, a structure-aware selector that grades detectors on synthetic probes built from the normal data, and it is the only method that significantly beats both survivors.

Our contributions:

1. **ADReal, a dataset and a protocol for evaluating anomaly detectors.** One detector-free, reproducible pipeline removes trivial, unsolvable, and mislabeled tasks, closes train/test leakage, caps the base rate, and deduplicates anomalies, so a detector's measured performance reflects skill rather than artifacts. From 173 constructed tasks, 151 survive.
2. **An evaluation of a rich set of modern detectors.** We fit roughly thirty classical and deep anomaly detectors under one consistent preprocessing, which gives each dataset its oracle-best detector.
3. **A protocol for evaluating unsupervised model selection, and an evaluation of the state of the art under it.** Beyond a detector's general performance, practice needs the best detector for a specific dataset given only its normal data, a different question. We score a selection method by its regret against the oracle-best detector, with its criterion computed on held-out normals only, and find that under this leakage-free protocol the agreement selectors (consensus, model-centrality, HITS) fall below a random pick while only entropy and mass-volume survive, reversing their apparent lead under transductive evaluation.
4. **SPARC, a model-selection method that beats the survivors.** SPARC builds synthetic anomalies from the normal data and grades detectors on how well they find them, and is the only method that significantly beats both entropy and mass-volume (regret 0.124; $p < 0.001$ against each), decisively on time series and holding on tabular data.

2. Related work

Anomaly detectors and benchmark suites. Established detectors span the standard families: isolation-based (Isolation Forest [8]), density- and neighbor-based (LOF [9], KNN), boundary-based (one-class SVM [10]), distribution-based (COPOD [11], ECOD [12], HBOS), projection-based (LODA [13], PCA), and deep one-class and reconstruction models (DeepSVDD [14] and successors [15]). Surveys [15] and large empirical studies agree that no detector dominates across datasets, which is the fact that makes per-dataset selection necessary. Benchmark suites standardize the comparison: PyOD [6] provides reference implementations; ADBench [5] consolidates 57 tabular datasets and 30 algorithms and reports that no method is statistically best; the DAMI study [16] curates the classic ODDS-derived tabular collection; MacroData [20] scales tabular outlier detection to thousands of datasets, including the OddBench (real-world semantic-anomaly tables) and OvrBench (one-vs-rest conversions of classification data) collections we draw on; and TSB-AD [19] curates over a thousand time-series under a reliable protocol. These resources standardize detectors and data, but not the protocol under which a label-free selection method is judged.

How detectors are scored, and why the metric matters. Tabular benchmarks report ROC-AUC, average precision (area under precision-recall), and precision-at-n; PyOD and DAMI [6, 16] lead with ROC-AUC and precision-at-n, ADBench [5] with ROC-AUC and AUC-PR. Under the heavy class imbalance of anomaly detection, precision-recall analysis is more informative than

ROC [26, 27], but average precision is not comparable across datasets of different anomaly rate: the precision-recall curve moves with prevalence [27], so the same detector scores differently as the base rate changes, and a raw AP gain can be a prevalence artifact rather than a real improvement [29]. Precision-recall-gain [28] rescales precision and recall against the always-positive baseline set by the base rate, which is the principle our metric follows. Time-series evaluation is more fragile still: the widely used point-adjustment protocol inflates scores so badly that a random score can look state of the art [31], and range-based and volume-under-the-surface measures [32, 33] were introduced to restore meaning. We take the lesson that a benchmark must fix one prevalence-aware metric rather than inherit these distortions.

Curation, preprocessing, and filtering in prior benchmarks. Data quality has been addressed before, piecemeal. The DAMI study [16] ships each dataset in normalized and unnormalized, with-duplicate and without-duplicate variants and in several downsampled contamination bands, and shows that normalization, duplicate removal, and outlier ratio each change method rankings, so one raw dataset yields many non-equivalent benchmark instances. Synthetic-anomaly benchmarks depend heavily on how anomalies are injected and only approximate genuine ones under strong assumptions [7]. TSB-AD [19] curates and filters time-series to drop unreliable cases. These works establish deduplication, contamination control, and filtering as necessary, but apply them locally and for detector comparison. ADReal turns them into one detector-free pipeline whose rules each close a specific contamination channel, applied uniformly across sources and aimed at both detector and selection-method evaluation.

Criticism of anomaly-detection benchmarks. A growing literature argues the benchmarks themselves manufacture progress. Wu and Keogh [30] show that popular time-series benchmarks are riddled with triviality, mislabeling, unrealistic anomaly density, and run-to-failure bias, so that a one-line moving average matches deep models. Röchner et al. [34] find that a trivial per-feature extreme-value rule is competitive with state-of-the-art deep detectors on standard tabular benchmarks, and argue evaluation must be rethought. These critiques target detector rankings; the same contamination is even more corrosive for selection, where a trivially separable or mislabeled task makes the choice among detectors meaningless. ADReal is built to remove exactly these channels.

Unsupervised model selection. Choosing a detector or its hyperparameters without labels is the harder problem, because both evaluation and comparison are

infeasible without ground truth [3]. Internal criteria score a detector from its own output: excess-mass and mass-volume curves [1] and the internal relative-evaluation score IREOS [2]. Meta-learning transfers performance from a labeled corpus of datasets: MetaOD [3] regresses dataset meta-features to expected performance, ELECT [21] adapts by dataset similarity under a budget, and HYPER [22] uses a hypernetwork to search deep-detector hyperparameters. Closest to our approach, a recent line grades detectors on synthetic anomalies generated from normal data: SWSA [24] builds a pseudo-validation set of generated image anomalies to rank detectors and prompts, and DSV [25] selects self-supervised augmentations by how well they align with the anomaly-generating mechanism. SPARC (Section 6) is in this family, for general tabular and time-series data.

Evaluating selection methods, and consensus. How selection methods are themselves evaluated is contested. A large-scale study of internal strategies [17, 23] concludes that no internal metric reliably beats a naive baseline, selecting models only comparable to a randomly configured detector. A common signal is agreement: averaging or combining detector scores yields a consensus that favors detectors agreeing with the majority [4, 18], and model-centrality and HITS-style rank aggregation formalize this within outlier ensembles [18]. Most of these methods are evaluated *transductively*, computing their criterion on the same data, anomalies included, that supplies the ground truth, which lets a selector's criterion see the very outliers it is asked to find. ADReal computes every selection criterion on held-out normals only and scores each method by regret against the oracle-best detector; the collapse of the agreement selectors we report in Section 5 is a direct consequence.

3. The ADReal benchmark

ADReal is produced by a single pipeline that never consults an anomaly detector (Figure 1). Every filtering and splitting decision is made from the data alone, so no design choice can be tuned to flatter a detector or a selection method. The pipeline curates raw source datasets into evaluation tasks and applies, in sequence, the contamination controls of Sections 3.2 to 3.5; two floors that depend on detector scores (the separability floor and the selection-trivial floor) are applied after the detector evaluation of Section 4, and the rest are detector-free. From 174 curated candidate tasks, 173 pass the construction gates and 151 remain after the detector-based floors.

Task structure. Each task is one dataset split three ways: a train set of normals, a validation set of normals,

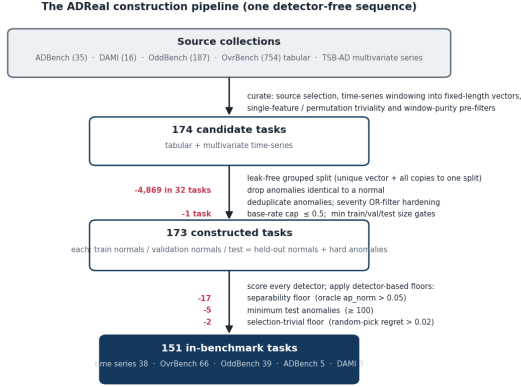


Figure 1. The ADReal construction pipeline. One detector-free sequence curates source collections into 174 candidate tasks, applies per-task contamination controls to reach 173 constructed tasks, and removes 22 tasks with three detector-based floors, leaving 151. Numbers in red are the tasks or anomalies removed at each step.

and a test set of held-out normals mixed with hard anomalies. This split carries the benchmark’s dual purpose. A detector is fit on the train normals; the test set, the only part that contains anomalies, scores detection quality; and the validation set contains normals only, so a label-free selection method calibrates its criterion exactly as it would in deployment, on normal data alone, without ever seeing an anomaly. No point appears in more than one split (Section 3.3).

3.1. Data sources

ADReal draws from two modalities. Tabular tasks come from ADBench [5], the DAMI collection [16], and the OddBench (real-world semantic-anomaly tables) and OvrBench (one-vs-rest conversions of classification data) collections of MacroData [20]; multivariate time-series tasks come from TSB-AD [19], windowed into fixed-length feature vectors. A curation stage selects usable source datasets, windows the series, and applies single-feature and permutation triviality pre-filters and a window-purity filter, leaving 174 candidate tasks that enter the pipeline of Figure 1. The final benchmark spans 151 tasks: 66 from OvrBench, 39 from OddBench, 38 time series, 5 from ADBench, and 3 from DAMI.

3.2. No trivially separable and no unsolvable anomalies

An anomaly a single feature or a one-line rule already isolates makes every detector look good and every selector’s choice irrelevant. We remove such anomalies with a calibrated OR rule that consults no detector: each feature is thresholded by its own empirical-tail and interior-gap-histogram rarity at a 5% false-positive rate, and a point is flagged if any feature fires, with the same rule repeated

in PCA-whitened space to catch oblique separations. For time series, a naive per-channel argmax fires on 56% of random label placements through multiple comparisons, so it is replaced by a permutation test against a circular-shift null at a 6.7% false-positive rate. At the opposite end, a task on which no detector beats a random ranking makes selection meaningless; a **separability floor** removes the 17 tasks whose oracle-best detector reaches base-rate-normalized average precision at or below 0.05, and a **selection-trivial floor** removes the 2 tasks on which every detector performs within a hair of the best, where any pick is a good pick.

3.3. No mislabeled anomalies and no leakage

Three failures let a selector be scored on the wrong answer. First, windowing mislabels: a sliding window inherits an anomaly flag from a single touched point, turning long stretches of normal signal into false anomalies. A stage-zero filter drops any series where fewer than half of anomaly points fall in contiguous segments of at least half a window. Second, label ambiguity: in coarse tabular data the same feature vector is sometimes labeled normal and sometimes anomalous, because the features do not capture what makes it anomalous; such an anomaly is an exact copy of a normal point and is unlabelable, so we remove every anomaly identical to a normal. Across the candidate tasks this removes 4,869 such anomalies in 32 tasks. Third, leakage: a naive random split of the normals leaks, because outlier collections contain many duplicate rows; on the source data it puts a normal into two splits in 62 of 173 tasks, 8,285 points in all, so a detector would be scored on rows it trained on. We instead split the normals 60/20/20 into train, validation, and test by a **grouped split** that assigns each unique feature-vector, together with all its duplicate copies, entirely to one split, so no point ever appears in two splits while the density of repeated values is preserved within each split; for time series a contiguous-block split plus purging of boundary-straddling windows serves the same end. Every label-free selection criterion is computed on the validation split, which contains normals only, so no selector observes a test anomaly.

3.4. One consistent KPI with a bounded base rate

Every detector is scored by a single quantity, base-rate-normalized average precision, $ap_norm = (AP - r) / (1 - r)$ where r is the task’s anomaly rate, comparable across tasks of different base rates. Because ap_norm divides by $1 - r$, a task whose test set is mostly anomalies makes the metric degenerate; a **base-rate cap** holds the test anomaly fraction at or below one half by subsampling hard anomalies to parity with the held-out normals, leaving a median test anomaly fraction of 0.35 across the

benchmark. A task must also retain enough anomalies to estimate the metric, so a floor of at least 100 test anomalies removes 5 further tasks. A selection method is then evaluated by its regret, the ap_norm of the oracle-best detector minus the ap_norm of the detector it picked.

3.5. No duplicated anomalies and one significance test

Repeated near-identical anomalies inflate whichever detector happens to catch the template. Datasets are deduplicated across sources by name and dimensionality, and within a task the anomaly set is reduced to distinct cases. Selection methods are compared by a paired Wilcoxon signed-rank test on per-task regret, reported two-sided, together with the median and tail of the regret distribution. A raw difference of means that a handful of easy tasks could produce is never reported as a result.

3.6. What the benchmark looks like

The 151 tasks are not uniform, and the two modalities differ in ways that matter for both detection and selection (Figure 2, Table 1). Tabular tasks range from 2 to over 300 features and sit near the base-rate cap, with a median test anomaly fraction of 0.43; their oracle detector is comparatively strong (median ap_norm 0.30). Time-series window features are higher-dimensional (median 16, up to 408), have a much lower prevalence (median 0.15), and are harder: the oracle detector reaches a median ap_norm of only 0.19. This spread is deliberate. The contamination controls remove the tasks on which selection would be meaningless, but leave a benchmark that is genuinely difficult, with an oracle far from perfect on most tasks, which is the regime in which a selection method’s choice actually matters.

Table 1. The 151 ADReal tasks by source: modality, number of tasks, and the median dimensionality, test anomaly fraction, and oracle-best detector ap_norm .

Source	Tasks	Dim (med.)	Anomaly frac. (med.)	Oracle ap_norm (med.)
OvrBench (tabular)	66	10	0.48	0.29
OddBench (tabular)	39	7	0.35	0.28
TSB-AD (time series)	38	16	0.15	0.19
ADBench (tabular)	5	36	0.50	0.51
DAMI (tabular)	3	21	0.23	0.65

4. Detectors and their evaluation

Detector pool. A selection method chooses from a fixed pool of 32 detectors that spans the classical families and a set of deep one-class and reconstruction models (Table 2). The classical part is the standard PyOD [6] set: Isolation Forest [8], LOF [9], KNN, ECOD [12], COPOD [11], HBOS, PCA, CBLOF, and LODA [13]. For the four algorithms whose single most influential hyperparameter drives their behavior, the pool sweeps a small grid of that hyperparameter rather than one default, so selection includes hyperparameter choice and not only algorithm choice: neighborhood size for LOF and KNN, histogram bins for HBOS, and retained variance for PCA. The deep part adds one detector each of DeepSVDD [14], an autoencoder, a variational autoencoder, internal contrastive learning [37], deep isolation, robust collaborative autoencoding, and scale learning [38]. Every detector, classical and deep, is fit on the train normals and scored on the same standardized features, so no detector is handicapped by a preprocessing inconsistency; the oracle-best detector per task defines the zero of regret.

Table 2. The 32-detector pool. For the four classical algorithms most sensitive to one hyperparameter, the pool sweeps a grid of that hyperparameter, so a selection method chooses configuration as well as algorithm.

Detector	Family	Hyperparameter swept	Variants
LOF [9]	Density / neighbor	neighborhood size k	6
KNN	Density / neighbor	neighborhood size k	6
HBOS	Distribution	histogram bins	4
PCA	Projection	retained variance	4
Isolation Forest [8], CBLOF	Isolation, density	default	2
ECOD [12], COPOD [11]	Distribution	default	2
LODA [13]	Projection	default (seeded)	1
DeepSVDD [14] and six more	Deep	one config each	7

No single detector dominates, and deep does not beat classical. The detector evaluation is the substrate for every selection result: on each task it fixes the oracle-best detector that defines the zero of regret, and no detector holds that role everywhere. Figure 3 ranks all thirty-two detectors by mean base-rate-normalized average precision across the 151 datasets. Neighbor and density methods (LOF and KNN variants) take the top six places; the best deep detector, DeepSVDD, is individually strong but ranks seventh by mean. Classical detectors supply the per-dataset oracle on 118 of 151 datasets (78%) against 33 for the deep pool, and the best deep detector

ADReal dataset characteristics (151 datasets): dimensionality, prevalence, and task difficulty

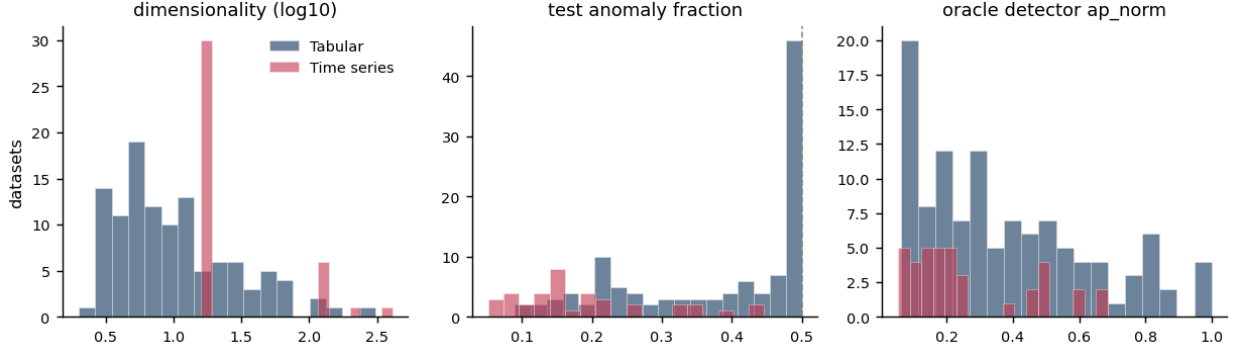


Figure 2. Characteristics of the 151 ADReal datasets, split by modality: dimensionality (log scale), test anomaly fraction (dashed line marks the base-rate cap at 0.5), and the difficulty of each task measured by its oracle-best detector. Time-series tasks are higher-dimensional, lower-prevalence, and harder than the tabular tasks.

per task trails the best classical by 0.071 on average, so adding the entire deep pool to the classical one raises the oracle by only 0.021. Fairly evaluated under one consistent preprocessing, classical detectors carry this benchmark, and none of the selection results below depend on the deep detectors.

Grouping detectors by family (Table 3) shows why no family is dispensable and why a family’s average is a poor guide to when it wins. The neighbor and density family supplies almost half of all per-dataset oracles. The distribution family has a low average yet supplies the oracle on nearly a quarter of datasets, because a few of its members win outright on specific datasets; the isolation family has a respectable average but is never the single best.

Table 3. Detector families on the 151-dataset ADReal benchmark: number of detectors in the pool, the share of datasets on which the family supplies the oracle-best detector, and the family’s mean base-rate-normalized average precision (ap_norm).

Family	Detectors	Oracle-best share	Mean ap_norm
Density / neighbor	13	45%	0.166
Distribution	6	24%	0.087
Deep	7	22%	0.123
Projection	5	9%	0.072
Isolation	1	0%	0.106

5. Unsupervised model selection: methods, protocol, and results

Selection methods. We evaluate eight label-free selectors, all restricted to the leakage-free protocol. *Entropy* (excess-mass) and *mass-volume* [1] score each detector by the shape of its score distribution over the data and a uniform reference sample.

Consensus [4], *model-centrality*, and *HITS* rank detectors by agreement with the pool [18]. *UDR* aggregates stability across configurations, *IFOREST-R* is a random-Isolation-Forest reference [17], and *IREOS* [2] scores the separability of the top-flagged points. Against these we place SPARC (Section 6). A uniform-random pick over the pool is the reference policy; a selector that fails to beat it carries no information.

Protocol. Every selection method receives the same inputs and is scored the same way. A detector is fit on the train normals; the method computes its criterion using only the validation normals and the detectors’ scores on them, with no access to any anomaly; it returns one chosen detector; and we score that choice by regret, the gap in test ap_norm between the oracle-best detector and the chosen one. Regret is aggregated over tasks and methods are compared by the paired Wilcoxon test of Section 3.5. Because the criterion never sees an anomaly, this measures label-free selection as it is actually deployed, unlike the transductive protocol under which these methods usually report gains.

Agreement-based selectors fall below a random pick. Figure 4 gives the selection regret of every method on the 151-task benchmark. The agreement-based methods that lead under transductive evaluation, consensus, model-centrality, and HITS, all land *above* the uniform-random reference (regret 0.232 to 0.235 versus 0.220), each significantly worse than SPARC ($p < 0.001$). With no anomalies in the validation split, the quantity these methods rank on is dominated by detector behavior on normals, and that quantity does not track detection skill. Entropy (EM) and mass-volume (MV) survive the protocol, at regret 0.176 and 0.174.

The collapse is a tabular effect. Breaking regret down by modality (Table 4) locates the failure. On the two tabular collections the agreement selectors reach regret 0.25 to 0.27 against a random pick of about 0.24, a clear collapse; on the time-series tasks the same methods sit near random (0.126 to 0.132 versus 0.135), neither collapsing nor helping. With

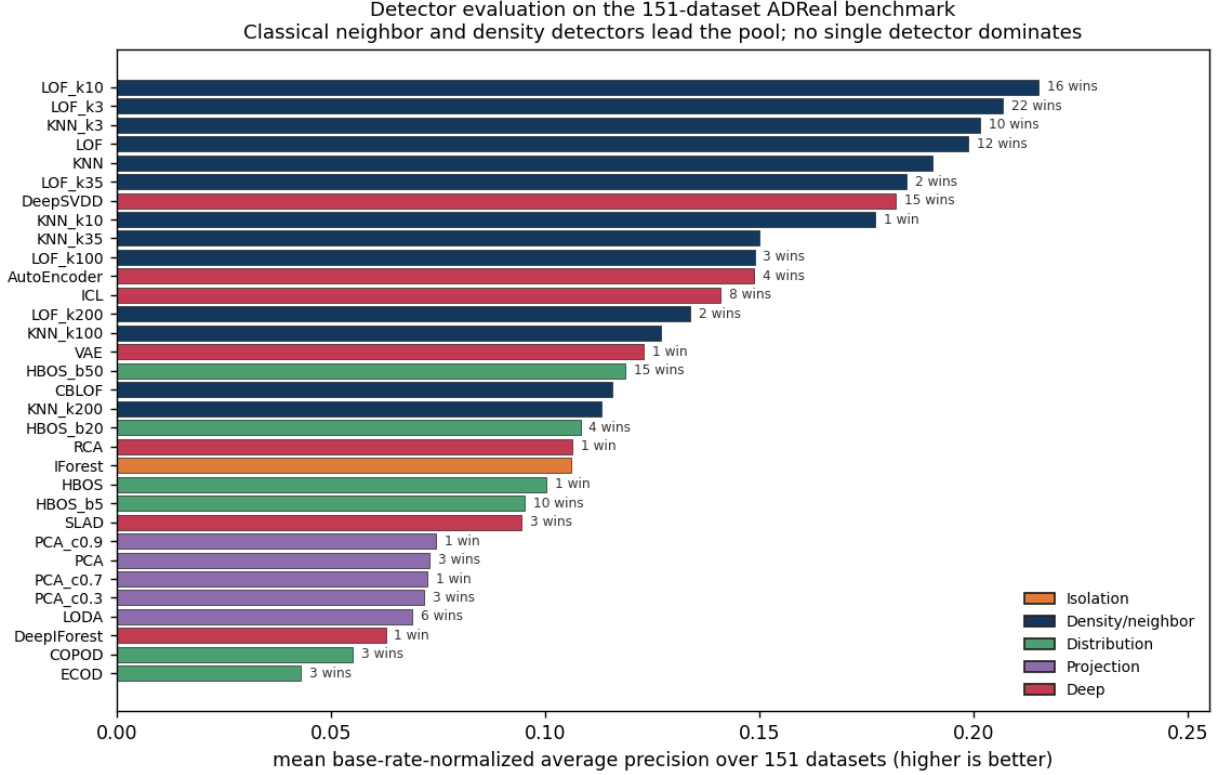


Figure 3. Detector evaluation on the 151-dataset ADReal benchmark: mean base-rate-normalized average precision per detector (higher is better), colored by family, with the number of datasets on which each detector is the oracle-best. Neighbor and density detectors lead; classical detectors supply the oracle on 78% of datasets, and the deep pool adds only 0.021 to the average oracle.

anomaly-free validation their criterion tracks how detectors behave on normals, which is actively misleading on the coarse tabular features and merely uninformative on the time-series windows. Entropy and mass-volume are the only baselines that beat random on both modalities, which is why they are the reference SPARC must clear.

Table 4. Selection regret (lower is better) by modality for all methods and the random reference. The agreement selectors (consensus, model-centrality, HITS) collapse on the tabular collections and sit near random on time series; only entropy, mass-volume, and SPARC beat random throughout.

Selector	All (151)	Time series (38)	OddBench (39)	OvrBench (66)
SPARC (Section 6)	0.124	0.054	0.151	0.144
Entropy (EM)	0.176	0.155	0.193	0.167
Mass-volume (MV)	0.174	0.169	0.180	0.168
Consensus	0.232	0.126	0.256	0.264
Model-centrality	0.233	0.126	0.262	0.263
HITS	0.235	0.132	0.251	0.269
Random pick	0.220	0.135	0.241	0.244

6. SPARC: selection by synthetic probes

Motivation. The surviving label-free criteria score a detector from the shape of its output on normal data alone; none of them ever confronts the detector with an anomaly. But a

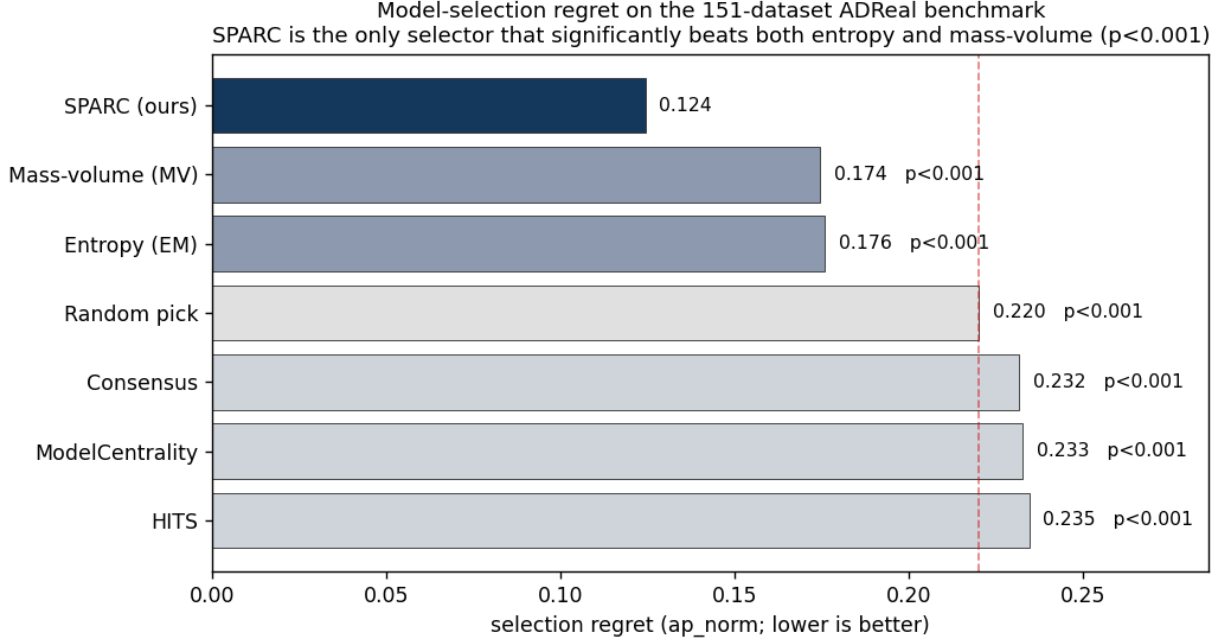


Figure 4. Model-selection regret on the 151-task ADReal benchmark (base-rate-normalized average precision; lower is better), with the two-sided paired Wilcoxon p -value against SPARC. Entropy and mass-volume survive the leakage-free protocol; the agreement-based selectors (consensus, model-centrality, HITS) fall at or above the random reference (dashed). SPARC is the only selector that significantly beats both survivors.

detector’s job is to separate anomalies from normals, and different detectors separate different kinds of anomaly. SPARC makes this concrete: it manufactures anomalies of known kinds from the normal data and asks each detector to find them, then selects the detector that does. It never observes a real anomaly, so it runs under the same normals-only protocol as the baselines. The idea of grading detectors on injected synthetic anomalies has been used for time-series selection before [35]; SPARC builds the anomalies to match the geometry of the normal data and routes between kinds, and applies to both tabular and time-series data.

Synthetic probes. From the validation normals, SPARC builds two kinds of synthetic anomaly, the two canonical ways a point can be an outlier [36]. A *dependency* probe resamples a fraction of each point’s features from their own marginal distributions, producing points whose per-feature values are ordinary but whose combination violates the normal correlation structure. A *marginal* probe pushes points into the tails of individual features. Each detector is graded by how well its scores separate each probe from the normals, giving a dependency grade and a marginal grade per detector.

Routing. SPARC routes between the best detector on each probe using a label-free property of the normal data: anisotropic datasets, whose covariance spectrum is concentrated in a few correlated directions, favor the dependency probe, while isotropic datasets favor the marginal probe; the two probe grades themselves supply a second, task-level signal of which probe is informative. Deep detectors tend to over-

fit the synthetic anomalies on coarse tabular data, separating the synthetic extremes without generalizing to the subtler real anomalies, so SPARC selects its probe winners among the classical detectors while the pool and the oracle still include the deep detectors.

Parameters and their effect. SPARC has three design choices, and Table 5 isolates each on the full benchmark. The *probe type* is set by a single generation parameter that controls whether a synthetic anomaly keeps ordinary per-feature values in an unusual combination (the dependency probe) or is pushed into the tails of individual features (the marginal probe). The dependency probe is the primary driver, reaching regret 0.122 on its own, and it carries the time series (0.054) and OvrBench (0.135); the marginal probe is weaker overall (0.145) but is the better probe on OddBench (0.147 versus 0.157), whose anomalies are coarse per-feature deviations. *Routing* between the two per task matches the stronger probe on every modality, giving SPARC its robustness across the two tabular collections at 0.124. The third choice, restricting the *probe candidates* to the classical detectors, is what significantly protects the method: allowing deep detectors to win a probe raises the marginal-probe regret from 0.145 to 0.166 ($p = 0.001$) and the routed regret to 0.143, because deep detectors overfit the synthetic extremes.

Table 5. SPARC ablation (mean regret over the 151 tasks, lower is better). The dependency probe is the primary driver; routing adds robustness across the tabular collections; restricting probe candidates to classical detectors is significant, since

deep detectors overfit the synthetic anomalies.

SPARC variant	Regret
SPARC (dependency + marginal, routed, classical candidates)	0.124
Dependency probe only	0.122
Marginal probe only	0.145
Routed, but deep detectors also eligible as probe winners	0.143

SPARC is the only selector that significantly beats both survivors. SPARC reaches regret 0.124, and the improvement over both survivors is significant at $p < 0.001$ against each (Figure 4). No other method clears both. The margin does not depend on the deep detectors: as the detector evaluation of Section 4 shows, a fairly preprocessed classical pool supplies almost every dataset’s oracle, so the selection comparison does not hinge on the deep models.

The advantage is decisive on time series and holds on tabular data. Figure 5 breaks the SPARC result down by modality. On time-series window features SPARC reaches regret 0.054 against entropy and 0.169 for mass-volume, a threefold reduction ($p < 0.001$). On tabular data SPARC beats entropy on the OddBench collection (0.151 versus 0.193, $p = 0.038$) and ties it on OvrBench (0.144 versus 0.167). The synthetic probes transfer cleanly to the structured window features of time series; on the coarse, low-cardinality features of the tabular semantic collections they are weaker proxies, and there entropy is already strong, so the two methods are matched.

7. Analysis and discussion

Where the synthetic probes work. SPARC’s selection quality follows the fidelity of its probes to real anomalies, and that fidelity follows the feature representation. Time-series window features carry rich dependency and marginal structure, so the synthetic probes rank detectors as the real anomalies do and SPARC selects near the oracle. The tabular semantic collections encode each record in a handful of coarse features that under-determine the label, so a synthetic anomaly is only a loose proxy for a real one; there the probes rank detectors weakly and SPARC gains no edge over entropy, which is already a strong criterion on that data. The benchmark thus locates precisely where label-free structure-aware selection helps and where no method has an edge.

Why honest measurement matters. The collapse of agreement-based selectors is not a small correction. Under a transductive protocol these methods can appear to lead; under the leakage-free protocol they fall below a random pick. The two protocols disagree about which methods work, and only one of them measures label-free selection as it is actually deployed. Constructing the benchmark also surfaced defects that quietly move results: anomalies identical to normal points, duplicate normals split across train and test, unbounded base rates, and

a preprocessing inconsistency that handicapped classical detectors and inflated the apparent value of deep ones. Each was removed, and the corrected benchmark reports a selection landscape in which a structure-aware method that reads the geometry of the normal data leads the field.

8. Limitations

ADReal covers tabular and multivariate time-series data; univariate series and other modalities are out of scope here, and the construction rules would need modality-specific instantiation to extend. SPARC’s advantage is decisive on time series and holds on tabular data, but on the coarse tabular semantic collections its synthetic probes are weak proxies and it only matches entropy; a probe construction that better fits low-cardinality features is the natural next step. The pool of detectors is finite, so the oracle that defines regret is the best *available* detector, not the best possible one.

9. Conclusion

Benchmarks for anomaly detection, and for the label-free methods that select among detectors, have been measuring artifacts: trivial and unsolvable tasks, anomalies that are copies of normals, leaked normals, degenerate base rates, and untested differences. ADReal rebuilds the measurement with a detector-free pipeline, audited until each of these channels is closed. On the clean benchmark it evaluates roughly thirty detectors under one consistent, prevalence-aware protocol, then evaluates selection methods by regret against the oracle-best detector under a leakage-free protocol. The honest protocol overturns the standing picture: agreement-based selectors fall below random, and SPARC, which grades detectors on label-free synthetic probes of the normal data, is the only selector that significantly beats both surviving baselines, decisively on time series and holding on tabular data.

Data and code availability

The benchmark construction pipeline, the per-task detector scores, the selection-method implementations, and the figures are released at the project repository. A permanent archive with a citable DOI accompanies the camera-ready version.

References

- [1] Goix, N. (2016). How to Evaluate the Quality of Unsupervised Anomaly Detection Algorithms? ICML Anomaly Detection Workshop . arXiv:1607.01152 . <https://arxiv.org/abs/1607.01152>
- [2] Marques, H. O., Campello, R. J. G. B., Zimek, A., Sander, J. (2015). On the Internal Evaluation of Unsupervised Outlier Detection. SSDBM . doi:10.1145/2791347.2791352 . <https://doi.org/10.1145/2791347.2791352>
- [3] Zhao, Y., Rossi, R. A., Akoglu, L. (2021). Automatic Unsupervised Outlier Model Selection. Advances in Neural Information Processing Systems 34 (NeurIPS) , pp. 4489-4502. arXiv:2009.10606 . <https://arxiv.org/abs/2009.10606>

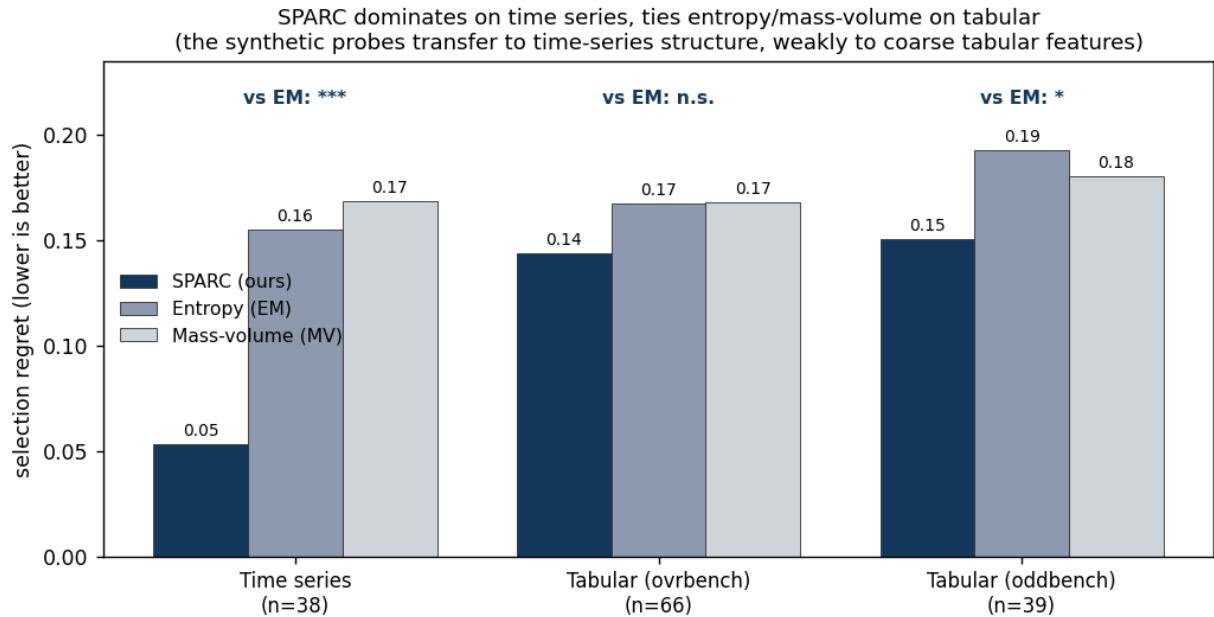


Figure 5. Selection regret by modality for SPARC versus the two surviving baselines. SPARC's advantage is decisive on time series ($p < 0.001$), significant on OddBench ($p = 0.038$), and a tie on OvrBench. Significance is the paired Wilcoxon test of SPARC against entropy.

- [4] Rayana, S., Akoglu, L. (2016). Less is More: Building Selective Anomaly Ensembles. ACM TKDD 10(4). doi:10.1145/2890508 . <https://doi.org/10.1145/2890508>
- [5] Han, S., Hu, X., Huang, H., Jiang, M., Zhao, Y. (2022). ADBench: Anomaly Detection Benchmark. NeurIPS Datasets and Benchmarks . arXiv:2206.09426 . <https://arxiv.org/abs/2206.09426>
- [6] Zhao, Y., Nasrullah, Z., Li, Z. (2019). PyOD: A Python Toolbox for Scalable Outlier Detection. JMLR 20(96). jmlr.org/papers/v20/19-011 . <https://jmlr.org/papers/v20/19-011.html>
- [7] Steinbuss, G., Böhm, K. (2021). Benchmarking Unsupervised Outlier Detection with Realistic Synthetic Data. ACM Transactions on Knowledge Discovery from Data 15(4), Article 65. doi:10.1145/3441453 . <https://doi.org/10.1145/3441453>
- [8] Liu, F. T., Ting, K. M., Zhou, Z.-H. (2008). Isolation Forest. IEEE International Conference on Data Mining (ICDM) , pp. 413-422. doi:10.1109/ICDM.2008.17 . <https://doi.org/10.1109/ICDM.2008.17>
- [9] Breunig, M. M., Kriegel, H.-P., Ng, R. T., Sander, J. (2000). LOF: Identifying Density-Based Local Outliers. ACM SIGMOD International Conference on Management of Data , pp. 93-104. doi:10.1145/342009.335388 . <https://doi.org/10.1145/342009.335388>
- [10] Schölkopf, B., Platt, J. C., Shawe-Taylor, J., Smola, A. J., Williamson, R. C. (2001). Estimating the Support of a High-Dimensional Distribution. Neural Computation 13(7), pp. 1443-1471. doi:10.1162/089976601750264965 . <https://doi.org/10.1162/089976601750264965>
- [11] Li, Z., Zhao, Y., Botta, N., Ionescu, C., Hu, X. (2020). COPOD: Copula-Based Outlier Detection. IEEE International Conference on Data Mining (ICDM) . arXiv:2009.09463 . <https://arxiv.org/abs/2009.09463>
- [12] Li, Z., Zhao, Y., Hu, X., Botta, N., Ionescu, C., Chen, G. H. (2022). ECOD: Unsupervised Outlier Detection Using Empirical Cumulative Distribution Functions. IEEE Transactions on Knowledge and Data Engineering . arXiv:2201.00382 . <https://arxiv.org/abs/2201.00382>
- [13] Pevný, T. (2016). Loda: Lightweight On-line Detector of Anomalies. Machine Learning 102(2), pp. 275-304. doi:10.1007/s10994-015-5521-0 . <https://doi.org/10.1007/s10994-015-5521-0>
- [14] Ruff, L., Vandermeulen, R. A., Görnitz, N., Deecke, L., Siddiqui, S. A., Binder, A., Müller, E., Kloft, M. (2018). Deep One-Class Classification. International Conference on Machine Learning (ICML) , PMLR 80, pp. 4393-4402. proceedings.mlr.press/v80/ruff18a . <https://proceedings.mlr.press/v80/ruff18a.html>
- [15] Ruff, L., Kauffmann, J. R., Vandermeulen, R. A., Montavon, G., Samek, W., Kloft, M., Dietterich, T. G., Müller, K.-R. (2021). A Unifying Review of Deep and Shallow Anomaly Detection. Proceedings of the IEEE 109(5). arXiv:2009.11732 . <https://arxiv.org/abs/2009.11732>

- [16] Campos, G. O., Zimek, A., Sander, J., Campello, R. J. G. B., Micenková, B., Schubert, E., Assent, I., Houle, M. E. (2016). On the Evaluation of Unsupervised Outlier Detection: Measures, Datasets, and an Empirical Study. *Data Mining and Knowledge Discovery* 30(4), pp. 891-927. doi:10.1007/s10618-015-0444-8 . <https://doi.org/10.1007/s10618-015-0444-8>
- [17] Ma, M. Q., Zhao, Y., Zhang, X., Akoglu, L. (2023). The Need for Unsupervised Outlier Model Selection: A Review and Evaluation of Internal Evaluation Strategies. *ACM SIGKDD Explorations* 25(1). doi:10.1145/3606274.3606277 . <https://doi.org/10.1145/3606274.3606277>
- [18] Aggarwal, C. C., Sathe, S. (2015). Theoretical Foundations and Algorithms for Outlier Ensembles. *ACM SIGKDD Explorations* 17(1), pp. 24-47. doi:10.1145/2830544.2830549 . <https://doi.org/10.1145/2830544.2830549>
- [19] Liu, Q., Paparrizos, J. (2024). TSB-AD: Towards a Reliable Time-Series Anomaly Detection Benchmark. *NeurIPS Datasets and Benchmarks* . arXiv:2412.20512 . <https://arxiv.org/abs/2412.20512>
- [20] Ding, X., Klüttermann, S., Wen, H., Chen, Y., Akoglu, L. (2026). MacROData: New Benchmarks of Thousands of Datasets for Tabular Outlier Detection. arXiv preprint . arXiv:2602.09329 . <https://arxiv.org/abs/2602.09329>
- [21] Zhao, Y., Zhang, S., Akoglu, L. (2022). Toward Unsupervised Outlier Model Selection. *IEEE International Conference on Data Mining (ICDM)* , pp. 773-782. arXiv:2211.01834 . <https://arxiv.org/abs/2211.01834>
- [22] Ding, X., Zhao, Y., Akoglu, L. (2024). Fast Unsupervised Deep Outlier Model Selection with Hypernetworks. *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)* . arXiv:2307.10529 . <https://arxiv.org/abs/2307.10529>
- [23] Ma, M. Q., Zhao, Y., Zhang, X., Akoglu, L. (2021). A Large-scale Study on Unsupervised Outlier Model Selection: Do Internal Strategies Suffice? arXiv preprint . arXiv:2104.01422 . <https://arxiv.org/abs/2104.01422>
- [24] Fung, C., Qiu, C., Li, A., Rudolph, M. (2024). Model Selection of Anomaly Detectors in the Absence of Labeled Validation Data. arXiv preprint . arXiv:2310.10461 . <https://arxiv.org/abs/2310.10461>
- [25] Yoo, J., Zhao, Y., Zhao, L., Akoglu, L. (2023). DSV: An Alignment Validation Loss for Self-supervised Outlier Model Selection. *ECML PKDD* . arXiv:2307.06534 . <https://arxiv.org/abs/2307.06534>
- [26] Davis, J., Goadrich, M. (2006). The Relationship Between Precision-Recall and ROC Curves. *International Conference on Machine Learning (ICML)* , pp. 233-240. doi:10.1145/1143844.1143874 . <https://doi.org/10.1145/1143844.1143874>
- [27] Saito, T., Rehmsmeier, M. (2015). The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. *PLoS ONE* 10(3): e0118432. doi:10.1371/journal.pone.0118432 . <https://doi.org/10.1371/journal.pone.0118432>
- [28] Flach, P., Kull, M. (2015). Precision-Recall-Gain Curves: PR Analysis Done Right. *Advances in Neural Information Processing Systems* 28 (NeurIPS) , pp. 838-846. proceedings.neurips.cc/paper/2015 . <https://proceedings.neurips.cc/paper/2015/hash/33e8075e9970de0cfea955afd4644bb2-Abstract.html>
- [29] McDermott, M., Hansen, L. H., Zhang, H., Gee, G., Kirchler, M., Kelly, C. J., Xu, N., Ghassemi, M. (2024). A Closer Look at AUROC and AUPRC under Class Imbalance. *Advances in Neural Information Processing Systems* 37 (NeurIPS) . arXiv:2401.06091 . <https://arxiv.org/abs/2401.06091>
- [30] Wu, R., Keogh, E. J. (2023). Current Time Series Anomaly Detection Benchmarks are Flawed and are Creating the Illusion of Progress. *IEEE Transactions on Knowledge and Data Engineering* 35(3), pp. 2421-2429. arXiv:2009.13807 . <https://arxiv.org/abs/2009.13807>
- [31] Kim, S., Choi, K., Choi, H.-S., Lee, B., Yoon, S. (2022). Towards a Rigorous Evaluation of Time-Series Anomaly Detection. *AAAI Conference on Artificial Intelligence* . arXiv:2109.05257 . <https://arxiv.org/abs/2109.05257>
- [32] Tatbul, N., Lee, T. J., Zdonik, S., Alam, M., Gottschlich, J. (2018). Precision and Recall for Time Series. *Advances in Neural Information Processing Systems* 31 (NeurIPS) . arXiv:1803.03639 . <https://arxiv.org/abs/1803.03639>
- [33] Paparrizos, J., Boniol, P., Palpanas, T., Tsay, R. S., Elmore, A., Franklin, M. J. (2022). Volume Under the Surface: A New Accuracy Evaluation Measure for Time-Series Anomaly Detection. *Proceedings of the VLDB Endowment* 15(11), pp. 2774-2787. doi:10.14778/3551793.3551830 . <https://doi.org/10.14778/3551793.3551830>
- [34] Röchner, P., Klüttermann, S., Rothlauf, F., Schlör, D. (2025). We Need to Rethink Benchmarking in Anomaly Detection. arXiv preprint . arXiv:2507.15584 . <https://arxiv.org/abs/2507.15584>
- [35] Goswami, M., Challu, C., Callot, L., Minorics, L., Kan, A. (2023). Unsupervised Model Selection for Time-series Anomaly Detection. *International Conference on Learning Representations (ICLR)* . arXiv:2210.01078 . <https://arxiv.org/abs/2210.01078>
- [36] Steinbuss, G., Böhm, K. (2021). Generating Artificial Outliers in the Absence of Genuine Ones: A Survey. *ACM Transactions on Knowledge Discovery from Data* 15(2), Article 30. arXiv:2006.03646 . <https://arxiv.org/abs/2006.03646>

[org/abs/2006.03646](https://arxiv.org/abs/2006.03646)

- [37] Shenkar, T., Wolf, L. (2022). Anomaly Detection for Tabular Data with Internal Contrastive Learning. International Conference on Learning Representations (ICLR). openreview.net/forum?id=_hszZbt46bT. https://openreview.net/forum?id=_hszZbt46bT
- [38] Xu, H., Wang, Y., Wei, J., Jian, S., Li, Y., Liu, N. (2023). Deep Anomaly Detection with Scale Learning. International Conference on Machine Learning (ICML). arXiv:2305.16114. <https://arxiv.org/abs/2305.16114>